

Studentkare — System Architecture & Features

End-to-End Application Architecture, User Workspaces, Features & AI Agent Mesh

Component / Area	Key Scope / Highlights	Target User / Role
Student Care Workspace	ABHA Health Vault, Marketplace, rPPG Vitals, Teleconsultation, Health Camp Day	STUDENT
Clinician EMR Console	Patient triage queue, EMR timeline, FHIR observations, clinical notes & CDSCO drug checks	NMC_DOCTOR
Vendor Partner Console	Diagnostic lab home collections, pharmacy delivery fulfillment, catalog pricing	VENDOR
Campus Admin Console	Hostel health velocity, early outbreak alerts, health camp station planning	CAMPUS_ADMIN
Super Admin Plane	Operations telemetry, Razorpay/Stripe billing, institutional contracts, VAVE Pen-Test gate	SUPER_ADMIN
AI Agent Mesh	9 Departmental Agents (D1-D9) + Crisis Gate, Outbreak Predictor, CDSCO & DPDP Reconciler	AUTONOMOUS

1. Role-Based Workspaces & User Flows

A. Student Care Workspace (/health, /shop, /digital-id, /records, /insurance)

- *Onboarding & Digital Health Card*: Phone/WhatsApp OTP authentication + campus roster matching. Generates an ABHA-linked QR card with zero-login emergency trauma profile.
- *Health Vault & Provenance*: Uploads lab reports and prescriptions with OCR LOINC/SNOMED extraction and provenance bounding-box highlights.
- *Marketplace*: Orders medicines, books lab test packages (pickup or home collection), and schedules teleconsultations.
- *Vitals & Prevention*: Smartphone camera rPPG vital signs scanning (heart rate, HRV, respiration) + Apple Health / Android Health Connect background sync.
- *Health Camp Day*: Station-by-station checkups (Dental, Vision, GP) to earn digital wellness badges.

B. Clinician EMR Console (/clinician, /clinical-notes, /report-reviews)

- Triage queue with chief complaints and urgency scores.
- Historical lab trend charts, FHIR observations, and OCR document verification.
- Clinical note authoring with CDSCO drug recall guards and drug interaction safety checks.

C. Vendor & Diagnostic Partner Console (/vendor)

- Order fulfillment tracking from sample collection through lab report upload.
- Real-time order state progression (*created* → *assigned* → *sample_collected* → *completed*).

D. Campus Admin & Super Admin Planes (/campus, /admin)

- Aggregate hostel health velocity monitoring and early viral outbreak warnings.
- Razorpay & Stripe subscription billing reconciliation, institutional seat contracts, and VAVE zero-trust pen-test safety logs.

2. The 9-Department AI Agent Mesh & Specialist Agents

StudentKare operates an autonomous AI system governed by a clinical constitution (*constitution.ts*):

Department / Agent	Autonomous Function & Responsibility
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D1 Service Desk	Automated ticket triage, issue escalation, and user support routing.
D2 Partner Ops	Diagnostic lab SLA monitoring and delivery partner fulfillment checks.
D3 Compliance	Enforces DPDP statutory erasure, CDSCO drug recalls, and FSSAI mess audits.
D4 Finance & Revenue	Subscription billing reconciliation, Razorpay/Stripe webhook verification, and seat limits.
D5 Claims Ops	M22–M25 insurance claim adjudication, fraud anomaly detection, and signature checks.
D6 Institution Success	University onboarding, student seat allocation, and aggregate telemetry reporting.
D7 Localization	Multi-lingual translation (English, Hindi, Telugu) and health guide adaptation.
D8 Engineering	Connection pool tracking, VAVE zero-trust gate enforcement, and error rate tracking.
D9 Clinical Governance	LOINC/SNOMED coding validation, drug-drug interaction gates, and allergy cross-checks.
Crisis Gate Agent	Detects self-harm keywords, locks AI response, and displays 24x7 crisis helplines.
Outbreak Predictor	Monitors hostel pyrexia velocity (R0 transmission estimate) to flag viral outbreaks.
CDSCO Recall Guard	Cross-checks prescribed drugs against active Drug Controller General India recall notices.
DPDP Erasure Agent	Reconciles data erasure requests across ABDM Gateway HIP/HIU nodes within 72h SLA.

3. Core Domain Entities & Data Structures

Entity	Key Attributes	Primary Usage
StudentProfile	id, fullName, phone, bloodGroup, abhaId, emergencyContact, allergies, pointsBalance	Digital Health Card & Student Portal
HealthRecord	id, category (LAB, PRESCRIPTION, VACCINE, CAMP_REPORT), documentUrl, fhirObservations, provenance	Health Vault & OCR Provenance
CatalogItem	id, name, pack, category (PRODUCT, LAB, CONSULTATION), pricePaise, mrpPaise, stock	Healthcare Marketplace
FabricOrder	id, partnerName, serviceCategory, modality (HOME_COLLECTION, PICKUP), state, providerCost	Health Services Fabric
ClaimAdjudication	id, policyNumber, claimantName, amountClaimedPaise, anomalyFlags, decisionStatus	Insurance & Claims Hub
HealthCamp	id, title, campusName, date, totalStations, completedCount, stations	Campus Health Camp Day